# Greedy algorithms exercises

## 16-E01 Activity selection and exchange

Difficulty: Basic

Activities(s,f) are(1,3),(2,5),(4,8),(7,9). Select maximum compatible activities by finish time and compare with choosing(2,5) first.

## 16-E02 Coin-greedy counterexample

Difficulty: Advanced

For denominations{1,4,5} and amount12, compare largest-denomination greedy with the true minimum coin count.

## 16-E03 Huffman merges

Difficulty: Advanced

Frequencies A4,B2,C1,D1. Trace merges, give an optimal prefix code, and compute weighted length.

## 16-E04 Weighted activities need DP

Difficulty: Research

short=[0,2),value1; long=[1,3),value100; tail=[3,4),value1. Compare earliest-finish greedy with weighted interval DP. Give recurrence and every prefix optimum.

## 16-E05 Augmentation does not follow from heredity

Difficulty: Research

S={a,b,c},weights10,9,9, I={∅,{a},{b},{c},{b,c}}. Prove heredity, disprove augmentation, and assess decreasing-weight greedy.

## 16-E06 Deadline capacities and a full schedule

Difficulty: Advanced

Unit jobs(name,deadline,penalty) are(r,1,12),(s,2,11),(t,1,10),(u,3,8),(v,2,7),all ready at0. Give N1,N2,N3 and feasibility at each penalty-greedy step, a full five-slot schedule, and minimum penalty. Prove general augmentation for n jobs.

## 16-E07 Huffman exchange sign and singleton boundary

Difficulty: Research

# Greedy algorithms exercises (continued)

Leaf x has frequency1,depth1; deepest leaf a has frequency5,depth3. Find the cost change when swapping them and derive its general sign. With alphabet{x} only, why do merge-weight sum and implementation code0 cost differ?

> Complete your derivations before opening the separate solutions PDF.

### References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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